

Real-World Real Estate

Pre-Sale with Proof & Release

FUNDACIÓN BLOCKCHAIN ARGENTINA

Project Catalyst

MILESTONE 4 · EVIDENCE 1

Pilot Report

Pilot execution on Cardano mainnet: metrics, successes, incidents and the notary's attestation on data integrity

Project Catalyst 1400106 · Real-World Real Estate: Pre-Sale with Proof & Release · October 31, 2026

Pilot Report

This document is the evidence for **Output 1** of Milestone 4: *"Pilot execution with one notary and two developers; real data; ≥2 releases per pilot; fallback/drill scenarios."* It reports the pilot's metrics for Criterion 1, its successes and incidents, and the notary's attestation on data integrity, as Evidence 1 asks. All figures were read from the mainnet deployment on **October 31, 2026** and checked against the public Cardano mainnet ledger.

The pilot ran a real project of Grupo Eagle Construcciones S.A. — Cabildo 4987, Av. Cabildo 4987, Núñez, Buenos Aires — end to end on Cardano mainnet: 10 construction stages certified, 63 real project documents fingerprinted and anchored in 15 evidence bundles, one unit sold under a signed contract, the unit's dossier signed by the notary, and the full contract amount recorded as released. Every one of its 48 transactions was verified on a public explorer, and every stored document re-hashes to the fingerprint anchored on-chain.

Scope

Developer	Grupo Eagle Construcciones S.A.
Project	Cabildo 4987 — 30 functional units, 10 construction stages, status *Delivered*
Data	The real documents of the project: plans, permits, certificates, surveys, site photographs
Network	Cardano mainnet (`https://propnexus-web-mainnet.onrender.com`)
Participants	The developer (Grupo Eagle Construcciones S.A.), a certifier, the notary, the buyer of unit UF 12 · 5°A, and the platform operator — one account each, every one bound to its role and to this project
Unit followed end to end	UF 12 · 5°A — 5th floor, 58 m², USD 146,100

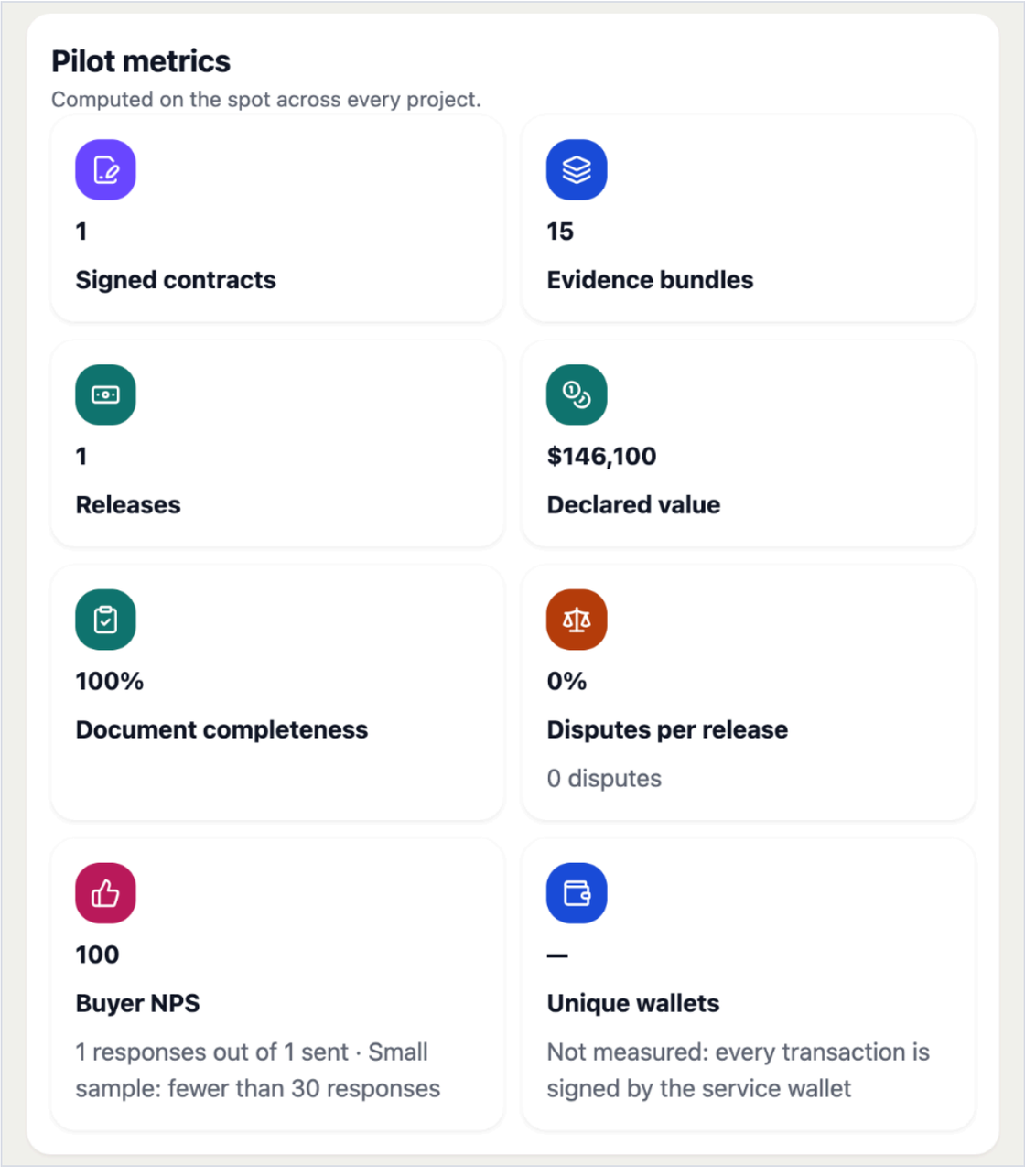
The pilot was run on **one project**, the one anchored on mainnet, and its results are reported as measured.

1 • Pilot metrics

Criterion 1 metric	Pilot result
Development pilots onboarded	**1** — Cabildo 4987, Grupo Eagle Construcciones S.A.
Signed contracts	**1** — unit UF 12 · 5°A, USD 146,100
Evidence bundles hashed	**15** — 63 documents
Releases	**1** — the full contract amount
Document completeness	**100 %**
Disputes	**0 %** — 0 disputes over 1 release
Buyer NPS	**100** — 1 response out of 1 sent
Unique wallets across pilots	**1** — every transaction is signed by the platform's service wallet
Value of the developments on the platform	**USD 3,665,414.93** — the 30 units of Cabildo 4987, priced in US dollars

How each figure is computed:

- **Document completeness** is the share of each unit's required artifacts that carry an on-chain reference. For the dossier of UF 12 · 5°A, 73 of 73 artifacts (10 stages and 63 documents) are anchored.
- **Disputes** are counted per release, so the rate exists only once there is a release.
- **NPS** is asked once per buyer, automatically, with their first release; the buyer answered 9. The platform flags any score based on fewer than 30 responses as a small sample.
- **Unique wallets** is not measured in this design: participants do not sign with their own wallets; the platform signs every anchor with its service wallet, and attributes each action to the identity-bound account that performed it.



The pilot metrics, as the admin panel of the mainnet deployment computes them.

2 • How the pilot ran

Timeline (October 31, 2026, UTC)

Time	What happened
17:54 – 18:06	Service wallet funded; reference script published on mainnet
18:43	Project created; its 10 stage threads opened on-chain (10 transactions)

Time	What happened
19:07 – 19:32	63 documents uploaded and fingerprinted, anchored in 15 evidence bundles
19:07 – 19:30	Work started on each stage (`Pending → InProgress`)
19:40 – 19:51	The certifier certified the 10 stages (`InProgress → Completed`)
20:21	The buyer accepted the invitation to unit UF 12 · 5ºA; contract signed
20:22	The notary signed the unit's dossier
21:33	The full contract amount recorded as released at stage 10; the buyer answered the NPS survey

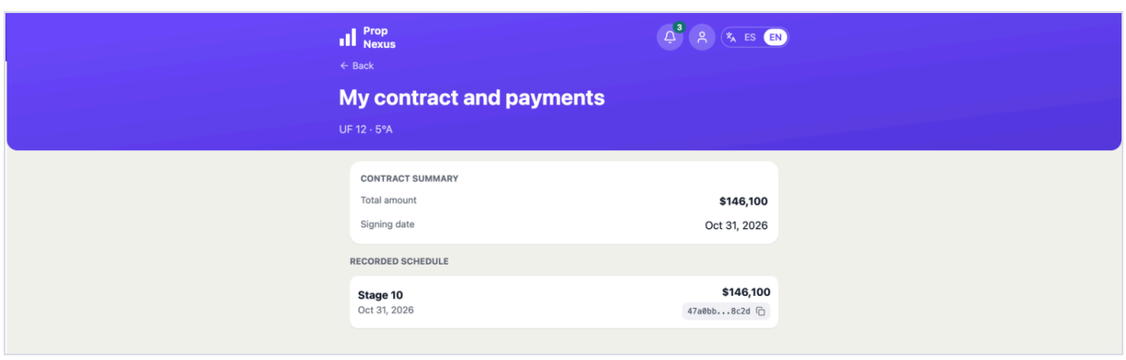
These timestamps record **when each artifact entered the record**, not when the construction work took place: the project's documents were registered on mainnet in a single session.

The ten stages

#	Stage (as recorded, es-AR)	English	Files	Work started (UTC)	Completed (UTC)	Completion transaction
1	Proyecto y Registro	Project and registration	2	19:08	19:40	`591273c7727ef9c8…`
2	Certificaciones de servicios y Estudio de suelos	Utility certifications and soil study	2	19:17	19:43	`dfad5a0ba41b4d93…`
3	Permiso de Ejecución de Obra Civil y comercialización	Civil works permit and sales authorization	3	19:18	19:43	`f8cb4f71b9ebf6bc…`
4	Inicio de obra y excavación	Construction start and excavation	3	19:20	19:46	`9a549bfec91123af…`
5	Fundaciones y estructura de planta baja	Foundations and ground-floor structure	11	19:20	19:47	`07ced4eea22021f9…`
6	Estructura de niveles superiores	Upper-floor structure	14	19:23	19:48	`262bb75c262b74c8…`
7	Instalaciones y terminaciones	Building services and finishes	2	19:26	19:49	`a6ea8cde9195a90f…`
8	Final de obra	Construction completion	4	19:28	19:50	`35fd5682499111d4…`
9	Mensura y División en Propiedad Horizontal	Survey and horizontal-property subdivision	2	19:29	19:51	`5988ab38e6301210…`
10	Escrituración por unidad funcional y entrega del inmueble finalizado	Deed per unit and handover of the finished property	20	19:30	19:52	`11a0a5a532f04a90…`

Releases

One release was recorded, for the full contract amount, at stage 10 (*Deed per unit and handover of the finished property*): USD 146,100.00, anchored in transaction `47a0bb3d…568c2d` . A release is the record of a payment made **outside** the platform — the platform never holds or moves funds.



The buyer's contract view, with the release recorded at stage 10 and its transaction ID.

Fallback and drill scenarios

In the mainnet pilot every stage of Cabildo 4987 was certified on its first review and no dispute was opened. Both fallback mechanisms are in production:

- **Failed inspection → rework → re-certification** is the `Observed ⇄ InProgress` branch of the stage state machine, enforced both by the backend and by the on-chain validator. It was exercised on 30 stages during the Milestone 3 volume test on Preprod (180 confirmed transactions).
- **Disputes** between the parties of a stage are recorded by the platform and counted in the dispute rate above.

3 · Successes

- **Real project, real documents, on mainnet.** A complete construction record of Grupo Eagle Construcciones S.A. — from the project registration to the deed and handover — was structured into 10 stages and anchored on Cardano mainnet.
- **Independent verification works.** Without relying on the platform's own word:
  - all **48** transaction IDs were found on mainnet through an independent API (Koios);
  - all **15** evidence-bundle Merkle roots were recomputed from the document hashes and match the roots on-chain;
  - all **63** stored documents (115.7 MB) were downloaded and hashed again, and every one reproduces the fingerprint that was anchored.
- **Certification rules held.** No stage could be completed without its evidence; document completeness is 100 %.
- **The professional output is usable.** The notary reviewed and signed the dossier of UF 12 · 5°A in the platform, and that signature is itself anchored and verifiable (§5).
- **No server errors.** Sentry shows no issues and Grafana shows no 5xx responses on mainnet (Evidence 2, §5).
- **Participants' feedback.** The participants reported that the platform works well, is satisfactory, and meets their requirements.

4 · Incidents

There were no incidents during the pilot: no data was lost or altered, and no anchor failed — all 48 transactions ended `Confirmed`.

Three defects were found while running the pilot, fixed the same day, and deployed through CI before the pilot ended. None affected the data:

Fix	What it was	Data impact
`c848681`	When several files were anchored together in one bundle, the screens looked the anchor up only for the first file, so the others showed *Pending* (seen in stage 1). The on-chain Merkle root always covered all the files	None — all 15 bundle roots verify against all 63 files
`cfc9f97`	Units were listed in text order (UF 1, UF 10, UF 11…) instead of by number	None — display only
`fe06df0`	The exported dossier PDF lost accented characters, `°` and `·`	None — export formatting only

5 · Notary attestation on data integrity

The notary attested to the integrity of the pilot's records **in the platform itself**, by signing the dossier of unit UF 12 · 5°A — and that signature is anchored on mainnet.

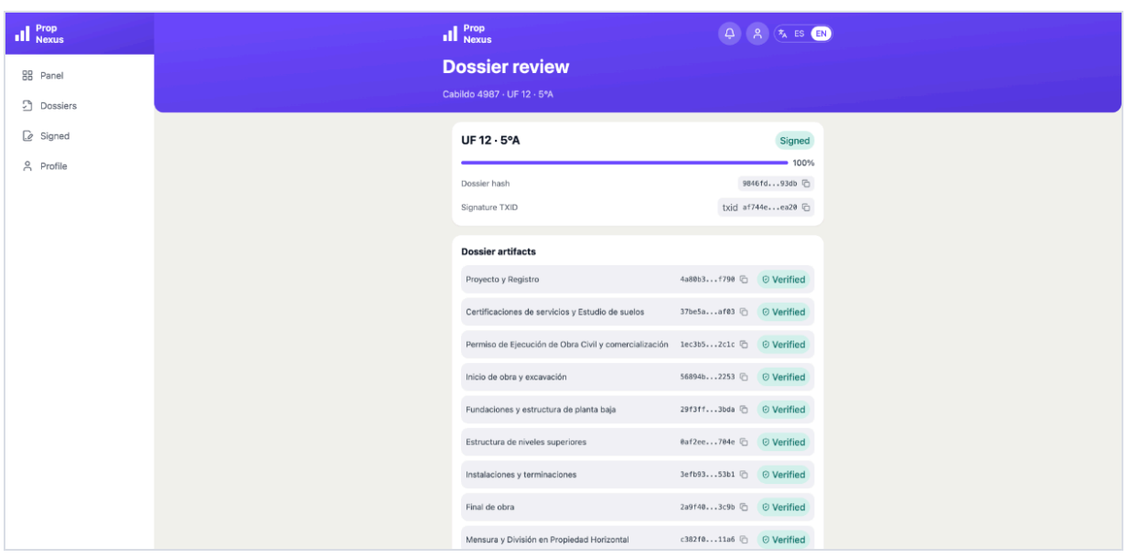
Dossier	`bitobzet1jkm3hzj39ufqx0j` — unit UF 12 · 5°A, Cabildo 4987
Contents	10 stages and 63 documents, each with its SHA-256 hash and anchor transaction
Master hash (what was signed)	`9846fd9b112416a1fb6cd5b4119e385198006685b659ce799f214016a0f393db`
Signed	October 31, 2026, 20:22:11 UTC, by the notary's account
Signature transaction	`af744e52c9ef141e61fd8c6da9609cf6f51f57e026bc06bfe7d778f1b376ea20`

How the attestation was verified.

1. The master hash was recomputed from the 73 artifacts of the dossier (10 stages and 63 documents, with their hashes and transactions): it is exactly `9846fd9b…93db`.
2. The commitment in the signature transaction's metadata was recomputed from the dossier ID, that master hash and the signing time: it is exactly the value on-chain, `4ed36503…9c8bfb`.
3. The 63 documents behind the dossier re-hash to their recorded fingerprints (§3).

The signature covers the dossier **as it was at 20:22:11 UTC**. The payment release recorded at 21:33 the same day is outside what the notary signed, and has its own transaction.

The Council of Experts format of the Whitepaper also calls for a written sign-off. The notary's sign-off on data integrity, which lists the same artifacts and states what the attestation does and does not cover, is submitted alongside this report as `M4-Sign-off-Notary-Data-Integrity.pdf`.



The signed dossier in the notary's view: status *Signed*, the master hash and the signature transaction.